

Concours Commun Mines-Ponts (CCMP) 2023

Physics I Exam, PC Track

Les sylphes

Updated Checked Solution

Contents

Checked answer key	3
1 Extracted figures	3
2 Observation des sylphes	7
Question 1	7
Question 2	7
Question 3	8
Question 4	8
Question 5	9
Question 6	9
Question 7	10
3 Une atmosphère à pression variable	10
Question 8	10
Question 9	11
Question 10	12
Question 11	13
Question 12	13
4 Une atmosphère électrique	14
Question 13	14
Question 14	14
Question 15	15
Question 16	15
Question 17	16

5 Formation avalancheuse de courant	16
Question 18	16
Question 19	17
Question 20	17
Question 21	18
Question 22	18
Question 23	19
Question 24	19
Final checked results	20

Checked answer key

Question	Main checked result
1	Sprites are red because the emission spectrum is strongest in the red / near infrared. Ground observation is hard because the downward path crosses absorbing atmosphere, especially the oxygen band near 762 nm.
2	Pic du Midi to the French Alps is of order 500 km; using a somewhat larger Mont Blanc distance, around 600 km, would give a horizon height closer to 30 km. In either case, sprites at 40 km to 80 km can clear the horizon, while Mont Blanc cannot in the no-refraction model.
3	The same localized signal appears in visible and in the 761 nm filtered image. A normal lightning flash would give a filtered/visible ratio $30/836 \simeq 3.6 \times 10^{-2}$, whereas the event has a much stronger relative narrow-band signal: this supports a sprite detection.
4–6	At 400 km: $G_{\text{ISS}} \simeq 9 \text{ m s}^{-2}$, $v_{\text{ISS}} \simeq 8 \text{ km s}^{-1}$, $T_{\text{ISS}} \simeq 90 \text{ min}$.
7	A 20 km camera field is crossed in about 2.5 s. A sprite lasting a few 0.1 s can be recorded, with roughly kilometre-scale precision.
8–12	Hydrostatic model gives $H = RT_0/(Mg_0) \simeq 9 \text{ km}$ and the pressure law in the statement. The model fails mainly because the assumed linear temperature law is unrealistic above the troposphere.
13–16	Collisionless plasma: $\gamma_p = -ine^2/(m_e\omega)$, $\omega^2 = \omega_p^2 + c^2k^2$, and for $n = 10^{11} \text{ m}^{-3}$, $f_p \simeq 3 \text{ MHz}$.
17	A 2.2 MHz radar reflects where $f_p \geq 2.2 \text{ MHz}$, i.e. $n_e \gtrsim 6 \times 10^{10} \text{ m}^{-3}$. The transient radar columns are consistent with sprite-related ionization.
18–20	$N(x) = N_0 \exp(-n_{\text{mol}}S_{\text{eff}}x)$, $\ell_{pm} = 1/(n_{\text{mol}}S_{\text{eff}})$, and at 60 km, $\ell_{pm} \simeq 2 \text{ cm}$.
21–24	A resting electron gains only about 2 eV over one mean free path in a 100 V m^{-1} field. A 1 MeV electron is relativistic. The avalanche stage number is $\mathcal{N} \simeq 35$, i.e. order 4×10^1 .

1 Extracted figures

These are the figures used in the reasoning below. They are included so the LaTeX file can be read without switching back to the exam statement.

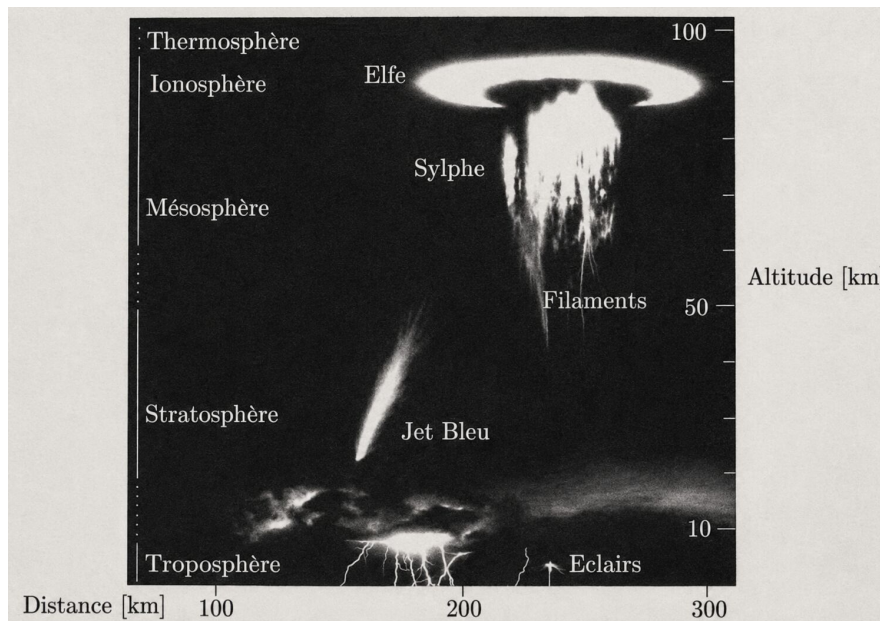


Figure 1: Figure 1: atmospheric luminous events and altitude scale.

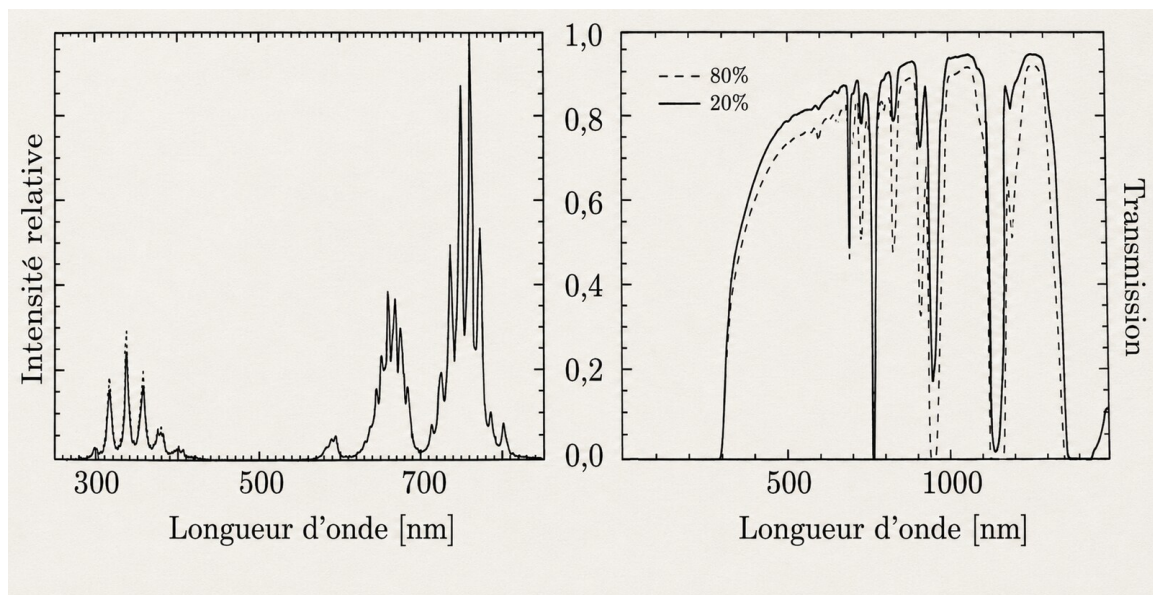


Figure 2: Figure 2: sprite emission spectrum and atmospheric transmission.

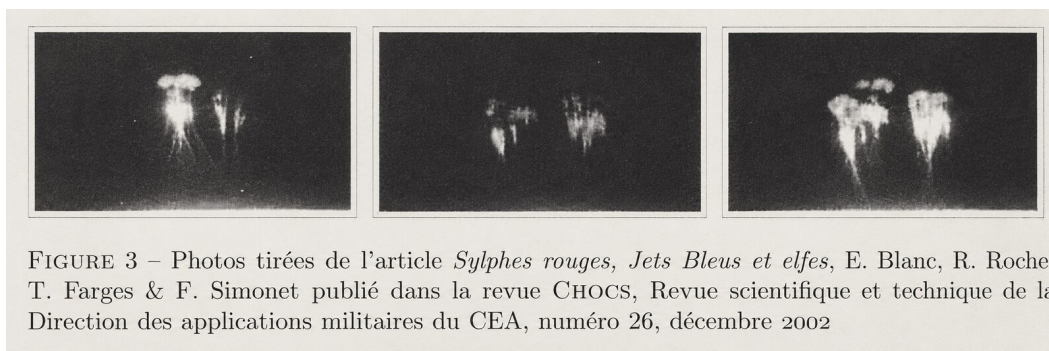


FIGURE 3 – Photos tirées de l'article *Sylphes rouges, Jets Bleus et elfes*, E. Blanc, R. Roche, T. Farges & F. Simonet publié dans la revue CHOCS, Revue scientifique et technique de la Direction des applications militaires du CEA, numéro 26, décembre 2002

Figure 3: Figure 3: sprite photographs taken from Pic du Midi.

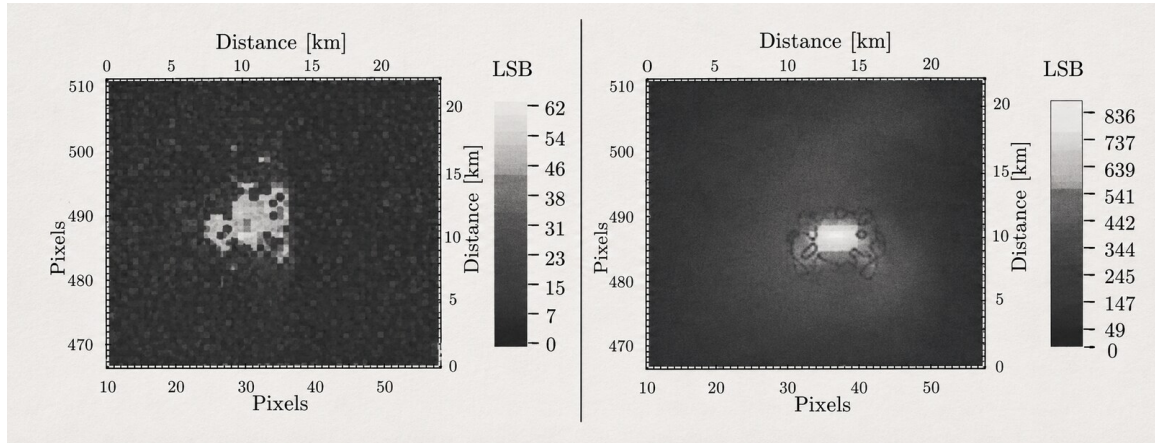


Figure 4: Figure 4: ISS filtered and visible microcamera images.

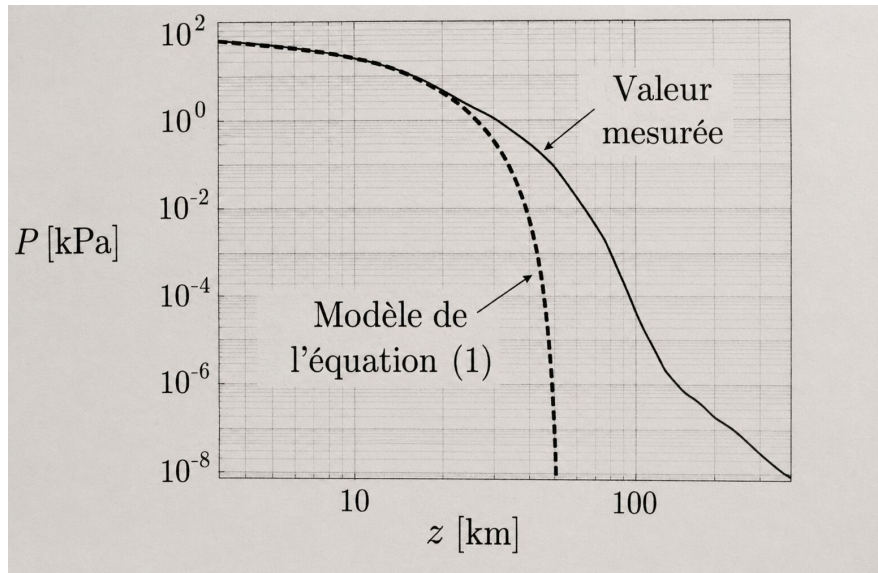


Figure 5: Figure 5: atmospheric pressure model compared with measured pressure.

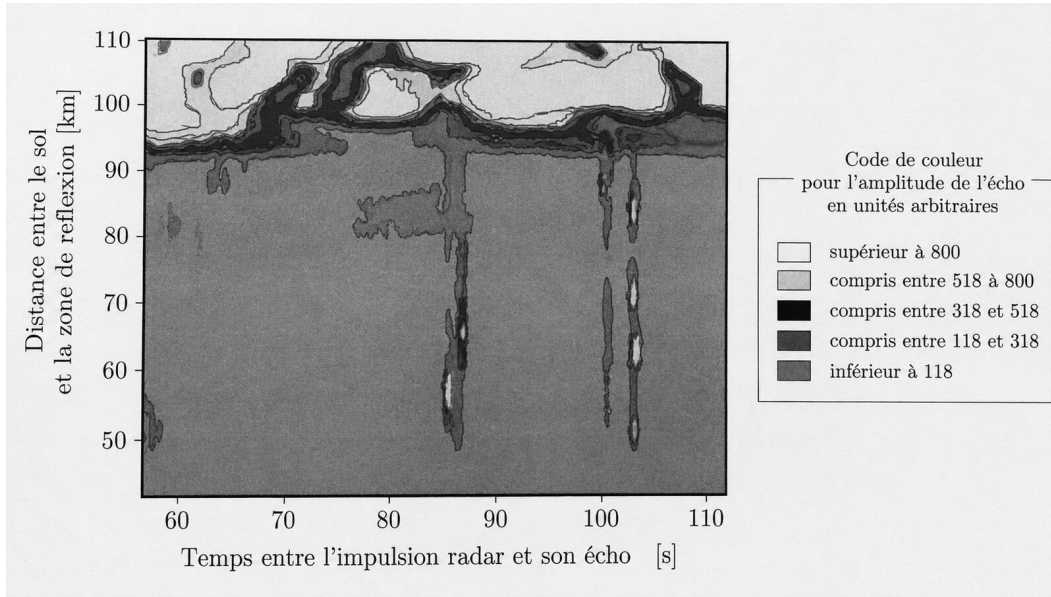


Figure 6: Figure 6: radar echo map at 2.2 MHz.

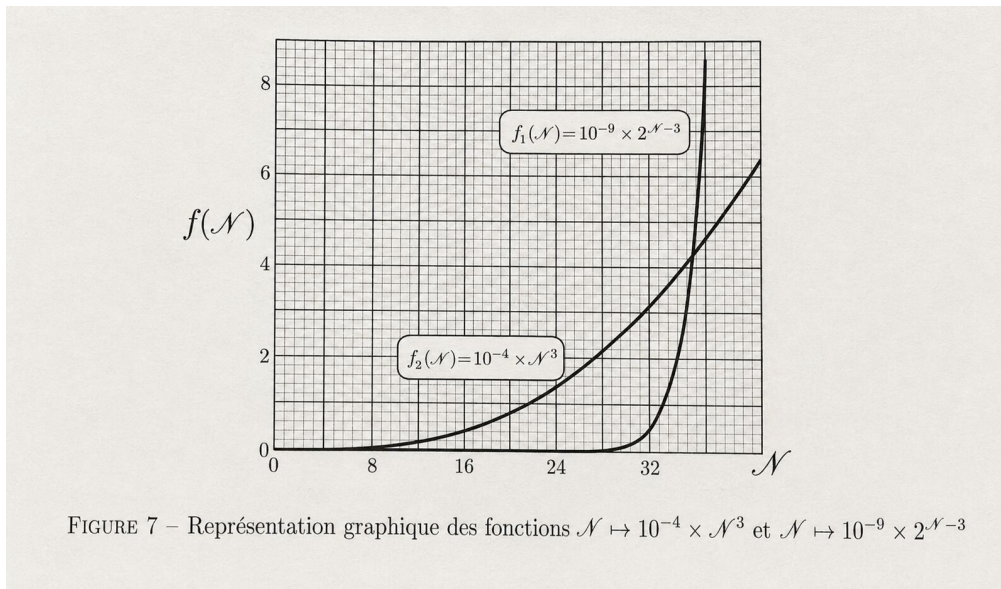
FIGURE 7 – Représentation graphique des fonctions $\mathcal{N} \mapsto 10^{-4} \times \mathcal{N}^3$ et $\mathcal{N} \mapsto 10^{-9} \times 2^{\mathcal{N}-3}$

Figure 7: Figure 7: graphical solution for the avalanche stage count.

2 Observation des sylphes

Question 1 – Colour and why ground observation is difficult

The left graph in Figure 2 is an emission spectrum. The largest peaks are mostly between about 600 nm and 800 nm. This is the red and near-infrared region of the spectrum. That is why sprites, or “sylphes”, have a red colour.

The right graph in Figure 2 shows atmospheric transmission for light emitted from a source at about 80 km altitude and detected from the ground. A transmission near 1 means the light passes through the atmosphere well; a transmission near 0 means the atmosphere absorbs it strongly. The curve has deep absorption dips, especially in the near infrared. The statement later points out the oxygen absorption band around 762 nm, very close to the sprite emission region. Water vapour also causes absorption, and the result depends on humidity.

So there are three reasons ground observation is difficult:

- 1) sprites are brief, lasting only a few hundred milliseconds;
- 2) they are far away and high in the atmosphere;
- 3) part of their red / near-infrared light is absorbed before reaching the ground.

Final answer. Sprites appear red because their strongest emission lies in the red / near-infrared. They are hard to observe from the ground because they are short-lived and because atmospheric absorption, especially near 762 nm, removes part of their light.

Question 2 – Earth curvature and visibility from Pic du Midi

The Pic du Midi is in the Pyrenees. The French Alps and Mont Blanc are several hundred kilometres away. Among 500 km, 1000 km, and 1500 km, the correct order of magnitude for the exam is therefore

$$d \sim 500 \text{ km.}$$

A more precise distance to a specific Alpine target such as Mont Blanc can be closer to about 600 km. This does not change the conclusion; it only raises the geometric horizon estimate from about 20 km to about 30 km.

Now ignore the altitude of Pic du Midi, as requested. The observer is placed on the surface of a spherical Earth. The line of sight to the horizon is tangent to Earth. A point at distance d along Earth’s surface is hidden below the tangent line by a height h .

Let $\theta = d/R_T$. From the right triangle formed by Earth’s centre, the observer, and the point above the horizon,

$$\cos \theta = \frac{R_T}{R_T + h}.$$

Therefore

$$h = R_T \left(\frac{1}{\cos(d/R_T)} - 1 \right).$$

For $d \ll R_T$, use $\cos \theta \simeq 1 - \theta^2/2$, so

$$h \simeq \frac{d^2}{2R_T}.$$

With $d = 500$ km and $R_T = 6400$ km,

$$h \simeq \frac{(5.0 \times 10^5 \text{ m})^2}{2 \times 6.4 \times 10^6 \text{ m}} \simeq 2.0 \times 10^4 \text{ m} \simeq 20 \text{ km}.$$

Sprites lie roughly between 40 km and 80 km, so they can be above this geometric horizon. Mont Blanc is only about 4.8 km high, which is far below even the 20 km to 30 km horizon-height estimate. In the no-refraction model, Mont Blanc is not visible from Pic du Midi at this distance.

Final answer. $d \sim 500$ km for the exam order of magnitude, giving $h \sim 20$ km. A 600 km distance would give $h \sim 30$ km. Alpine sprites can be visible because they are much higher than this, but Mont Blanc cannot be seen in this simplified no-refraction geometry.

Question 3 – Why Figure 4 proves a sprite detection

Figure 4 compares two simultaneous ISS images: a narrow-band image filtered around 761 nm and a visible-light image. The statement gives a useful calibration: a normal lightning flash gives about 836 LSB in the visible camera but only about 30 LSB in the filtered camera. For ordinary lightning, the filtered/visible ratio is therefore only

$$\frac{30}{836} \simeq 3.6 \times 10^{-2}.$$

In the event shown, the same localized object appears in the two images, at the same place. The filtered image has a clear signal in the narrow band associated with sprite emissions. Reading the colour bars only approximately, the filtered image reaches tens of LSB while the visible image reaches a few hundred LSB, so the relative narrow-band signal is much larger than the ordinary-lightning calibration ratio. This is important because random noise would not appear at the same position in both cameras, and ordinary lightning would not be expected to give such a strong narrow-band signature.

Final answer. The co-located visible and 761 nm signals identify the same luminous event, and the filtered/visible ratio is much larger than the $30/836 \simeq 0.036$ ordinary-lightning calibration. This supports the detection of a sprite.

Question 4 – Gravity at ISS altitude

The ISS is at altitude $h_{\text{ISS}} \simeq 400$ km. Outside a spherically symmetric Earth, Gauss' theorem tells us that the gravitational field is the same as if all of Earth's mass were concentrated at its centre.

At Earth's surface,

$$g_0 = \frac{GM_T}{R_T^2}.$$

At ISS altitude,

$$G_{\text{ISS}} = \frac{GM_T}{(R_T + h_{\text{ISS}})^2}.$$

Divide the second equation by the first:

$$G_{\text{ISS}} = g_0 \left(\frac{R_T}{R_T + h_{\text{ISS}}} \right)^2 = g_0 \left(1 + \frac{h_{\text{ISS}}}{R_T} \right)^{-2}.$$

Numerically,

$$\frac{h_{\text{ISS}}}{R_T} = \frac{400}{6400} = 0.0625,$$

so

$$G_{\text{ISS}} = \frac{9.8}{(1.0625)^2} \simeq 8.7 \text{ m s}^{-2}.$$

Final answer.

$$G_{\text{ISS}} = g_0 \left(1 + \frac{h_{\text{ISS}}}{R_T} \right)^{-2} \simeq 9 \text{ m s}^{-2}.$$

Question 5 – ISS orbital speed

For a circular orbit, the speed is constant in magnitude, but the velocity direction changes. The required centripetal acceleration is

$$a_c = \frac{v_{\text{ISS}}^2}{R_T + h_{\text{ISS}}}.$$

Gravity provides this acceleration, so

$$\frac{v_{\text{ISS}}^2}{R_T + h_{\text{ISS}}} = G_{\text{ISS}}.$$

Thus

$$v_{\text{ISS}} = \sqrt{G_{\text{ISS}}(R_T + h_{\text{ISS}})}.$$

Using the expression from Question 4,

$$v_{\text{ISS}} = \sqrt{\frac{g_0 R_T^2}{R_T + h_{\text{ISS}}}}.$$

Numerically,

$$v_{\text{ISS}} \simeq \sqrt{\frac{9.8(6.4 \times 10^6 \text{ m})^2}{6.8 \times 10^6 \text{ m}}} \simeq 7.7 \times 10^3 \text{ m s}^{-1}.$$

Final answer.

$$v_{\text{ISS}} \simeq 8 \text{ km s}^{-1}.$$

Question 6 – ISS period and Kepler's third law

One circular orbit has length $2\pi(R_T + h_{\text{ISS}})$. Therefore

$$T_{\text{ISS}} = \frac{2\pi(R_T + h_{\text{ISS}})}{v_{\text{ISS}}}.$$

Substitute the speed from Question 5:

$$T_{\text{ISS}} = 2\pi \sqrt{\frac{(R_T + h_{\text{ISS}})^3}{g_0 R_T^2}}.$$

Since $g_0 R_T^2 = GM_T$, this is also

$$T_{\text{ISS}} = 2\pi \sqrt{\frac{r^3}{GM_T}}, \quad r = R_T + h_{\text{ISS}}.$$

Squaring,

$$T_{\text{ISS}}^2 = \frac{4\pi^2}{GM_T} r^3.$$

This is Kepler's third law: T^2 is proportional to r^3 .

Numerically,

$$T_{\text{ISS}} \simeq \frac{2\pi(6.8 \times 10^6 \text{ m})}{7.7 \times 10^3 \text{ m s}^{-1}} \simeq 5.6 \times 10^3 \text{ s} \simeq 93 \text{ min}.$$

Final answer.

$$T_{\text{ISS}} \simeq 90 \text{ min}, \quad T^2 \propto r^3.$$

Question 7 – Can the ISS cameras record a whole sprite?

Figure 4 has a field of view of about 20 km by 20 km. With $v_{\text{ISS}} \simeq 8 \text{ km s}^{-1}$, the time for the ISS ground track to cross 20 km is

$$\Delta t_{\text{field}} \simeq \frac{20 \text{ km}}{8 \text{ km s}^{-1}} \simeq 2.5 \text{ s}.$$

A sprite lasts only a few hundred milliseconds. For example,

$$0.1 \text{ s} \Rightarrow \Delta x \simeq 0.8 \text{ km}, \quad 0.3 \text{ s} \Rightarrow \Delta x \simeq 2.4 \text{ km}.$$

These distances are much smaller than the 20 km field of view, so recording the whole event is possible if the storm lies inside the field at the right time.

For precision, the image scale is roughly

$$\frac{20 \text{ km}}{40\text{--}50 \text{ pixels}} \simeq 0.4 \text{ km/pixel to } 0.5 \text{ km/pixel}.$$

The ISS also moves by about a kilometre during a typical event. So the natural spatial precision is not metres; it is about a kilometre.

Final answer. Recording a whole sprite is possible: the field is crossed in about 2.5 s, while the sprite lasts only a few 0.1 s. The spatial precision is roughly kilometre-scale because of the pixel size and ISS motion.

3 Une atmosphère à pression variable

Question 8 – Hydrostatic equation and scale height

The vector law of hydrostatic equilibrium is

$$\text{grad } P = \mu \mathbf{g},$$

where μ is the mass density and $\mathbf{g}$ points downward. If z is upward, this becomes

$$\frac{dP}{dz} = -\mu g(z).$$

The atmosphere is treated as an ideal gas. For molar mass M ,

$$\mu = \frac{PM}{RT(z)}.$$

The temperature model is

$$T(z) = T_0 \left(1 - \frac{z}{\ell}\right),$$

and gravity is

$$g(z) = g_0 \left(1 + \frac{z}{R_T}\right)^{-2}.$$

Therefore

$$\frac{dP}{dz} = -\frac{PM}{RT_0(1 - z/\ell)} g_0 \left(1 + \frac{z}{R_T}\right)^{-2}.$$

Define the characteristic height

$$H = \frac{RT_0}{Mg_0}.$$

Then

$$\frac{dP}{dz} = -F(z) \frac{P}{H}, \quad F(z) = \frac{1}{(1 - z/\ell)(1 + z/R_T)^2}.$$

Numerically, with $T_0 = 290$ K and $M = 29 \times 10^{-3}$ kg mol⁻¹,

$$H = \frac{8.31 \times 290}{0.029 \times 9.8} \simeq 8.5 \times 10^3 \text{ m} \simeq 9 \text{ km}.$$

Final answer.

$$\frac{dP}{dz} = -F(z) \frac{P}{H}, \quad F(z) = \frac{1}{(1 - z/\ell)(1 + z/R_T)^2}, \quad H = \frac{RT_0}{Mg_0} \simeq 9 \text{ km}.$$

Question 9 – Pressure law

The statement gives a decomposition

$$F(z) = \frac{\alpha}{1 - z/\ell} + \frac{\beta + \gamma z}{(1 + z/R_T)^2},$$

with $\varepsilon = \ell/R_T$. Since

$$\varepsilon = \frac{50}{6400} \simeq 8 \times 10^{-3} \ll 1,$$

we may use

$$\alpha \simeq 1, \quad \beta \simeq 2\varepsilon, \quad \gamma \simeq \frac{\varepsilon}{R_T}.$$

The pressure equation is

$$\frac{dP}{P} = -\frac{F(z)}{H} dz.$$

Integrate from 0 to z , using $P(0) = P_0$.

First term:

$$\int_0^z \frac{du}{1 - u/\ell} = -\ell \ln \left(1 - \frac{z}{\ell} \right).$$

This gives the factor

$$\left(1 - \frac{z}{\ell} \right)^{\ell/H}.$$

For the second term, with $y = 1 + u/R_T$, one obtains

$$\int_0^z \frac{2\varepsilon + (\varepsilon/R_T)u}{(1 + u/R_T)^2} du = \ell \left[\ln \left(1 + \frac{z}{R_T} \right) + 1 - \frac{1}{1 + z/R_T} \right].$$

Here we used $\varepsilon R_T = \ell$. Also,

$$1 - \frac{1}{1 + z/R_T} = \frac{z/R_T}{1 + z/R_T}.$$

So, after multiplication by $-1/H$, the non-logarithmic part contributes

$$-\frac{\ell}{H} \frac{z/R_T}{1 + z/R_T} = -\frac{z}{(HR_T/\ell)(1 + z/R_T)}.$$

This gives both a denominator power of $1 + z/R_T$ and a small exponential correction.

Thus

$$P(z) = P_0 \left(\frac{1 - z/\ell}{1 + z/R_T} \right)^{\xi} \exp \left[-\frac{z}{\chi(1 + z/R_T)} \right],$$

where

$$\xi = \frac{\ell}{H}, \quad \chi = \frac{HR_T}{\ell}.$$

Numerically,

$$\xi \simeq \frac{50}{8.5} \simeq 5.9 \simeq 6, \quad \chi \simeq \frac{8.5 \times 6400}{50} \text{ km} \simeq 1.1 \times 10^3 \text{ km}.$$

Final answer.

$$P(z) = P_0 \left(\frac{1 - z/\ell}{1 + z/R_T} \right)^{\ell/H} \exp \left[-\frac{z}{(HR_T/\ell)(1 + z/R_T)} \right].$$

So $\xi = \ell/H$ and $\chi = HR_T/\ell$.

Question 10 – Useful simplification of the pressure law

For altitudes below 50 km, the ratio z/R_T is tiny:

$$\frac{50}{6400} \simeq 8 \times 10^{-3}.$$

So

$$1 + \frac{z}{R_T} \simeq 1.$$

Also $\chi \simeq 1100$ km, so

$$\exp \left[-\frac{z}{\chi(1 + z/R_T)} \right] \simeq 1$$

for $z \leq 50$ km.

Therefore the practical simplified pressure law is

$$P(z) \simeq P_0 \left(1 - \frac{z}{\ell}\right)^{\ell/H}.$$

This simplification amounts to neglecting the variation of gravity with altitude while keeping the imposed temperature law.

At very low altitude, if $z \ll \ell$, one can go further:

$$\left(1 - \frac{z}{\ell}\right)^{\ell/H} \simeq \exp\left(-\frac{z}{H}\right),$$

which is the familiar exponential barometric law.

Final answer.

$$P(z) \simeq P_0 \left(1 - \frac{z}{\ell}\right)^{\ell/H}.$$

This keeps the temperature decrease but neglects Earth-curvature / gravity variation.

Question 11 – Why the model fails near 20 km

The model was built with the temperature law

$$T(z) = T_0 \left(1 - \frac{z}{\ell}\right), \quad \ell = 50 \text{ km}.$$

This is a simple linear decrease. It is acceptable only as a rough lower-atmosphere model. The real atmosphere does not keep cooling linearly up to 50 km: near the tropopause, around 10 km to 20 km, the temperature stops following the tropospheric decrease, and in the stratosphere it can even increase.

If the model makes the temperature too low, then the density and pressure decrease too quickly. This matches Figure 5: the model pressure drops away from the measured curve above about 20 km.

Final answer. The main failure is the unrealistic linear temperature profile. It stops being valid around the tropopause / lower stratosphere, so the predicted pressure becomes too small above about 20 km.

Question 12 – Observation from Earth and from the ISS

The oxygen molar fraction is assumed constant:

$$x_{\text{O}_2} \simeq 0.20.$$

Therefore the oxygen partial pressure is

$$P_{\text{O}_2}(z) = x_{\text{O}_2} P(z) \simeq 0.20 P(z).$$

Near the ground, $P(z)$ is large, so P_{O_2} is large. Light travelling downward from a sprite to a ground observer must cross dense oxygen-rich layers. Around 762 nm, molecular oxygen absorbs strongly, so the signal can be weakened.

From the ISS, the light path goes upward from the sprite. It avoids most of the dense lower atmosphere. At altitudes of 40 km to 80 km, the pressure and oxygen partial pressure are already much smaller than at the ground. Absorption is therefore much weaker for upward observation.

Final answer. Ground observation is difficult because the light crosses dense oxygen-rich air, especially near the 762 nm absorption band. ISS observation is easier because the line of sight avoids most of the absorbing lower atmosphere.

4 Une atmosphère électrique

Question 13 – Neglecting the magnetic Lorentz force

The Lorentz force on an electron is

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}).$$

In an electromagnetic wave,

$$B \sim \frac{E}{c}.$$

So the ratio of the magnetic part to the electric part is

$$\frac{|q\mathbf{v} \times \mathbf{B}|}{|q\mathbf{E}|} \sim \frac{vB}{E} \sim \frac{v}{c}.$$

The problem says the electron motion is non-relativistic, so $v \ll c$.

Final answer. The magnetic force is smaller than the electric force by a factor $v/c \ll 1$, so it is negligible.

Question 14 – Electron current and plasma conductivity

Ions are much heavier than electrons. Under the same electric field, their acceleration is much smaller. Therefore their contribution to the current is negligible compared with the electron contribution.

Let $e > 0$ be the elementary charge; an electron has charge $-e$. The equation of motion is

$$m_e \frac{d\mathbf{v}}{dt} = -e\mathbf{E}.$$

The field has complex time dependence $\exp(i\omega t)$, so d/dt becomes $i\omega$:

$$i\omega m_e \mathbf{v} = -e\mathbf{E}.$$

Therefore

$$\mathbf{v} = \frac{ie}{m_e \omega} \mathbf{E}.$$

The electron current density is

$$\mathbf{j} = n(-e)\mathbf{v}.$$

Thus

$$\mathbf{j} = -i \frac{ne^2}{m_e \omega} \mathbf{E}.$$

By definition, $\mathbf{j} = \gamma_p \mathbf{E}$, so

$$\gamma_p = -i \frac{ne^2}{m_e \omega}.$$

This conductivity is purely imaginary. The average power density transferred to matter is proportional to $\text{Re}(\gamma_p)$, which is zero here. With no collisions, the plasma does not dissipate energy on average; it exchanges energy reversibly with the wave.

Final answer.

$$\gamma_p = -i \frac{ne^2}{m_e \omega}.$$

It is purely imaginary, so the average absorbed power is zero in this collisionless model.

Question 15 – Wave equation and dispersion relation

Use Maxwell's equations in a plasma with current density $\mathbf{j}$:

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{j} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}, \quad \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}.$$

Taking the curl of Faraday's law and using $\nabla \cdot \mathbf{E} = 0$ for a transverse plane wave gives

$$\Delta \mathbf{E} - \mu_0 \varepsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2} = \mu_0 \frac{\partial \mathbf{j}}{\partial t}.$$

Since $c^2 = 1/(\mu_0 \varepsilon_0)$,

$$\Delta \mathbf{E} - \frac{1}{c^2} \frac{\partial^2 \mathbf{E}}{\partial t^2} = \mu_0 \frac{\partial \mathbf{j}}{\partial t}.$$

Insert

$$\mathbf{E} = \mathbf{E}_0 e^{i(\omega t - k z)}, \quad \mathbf{j} = \gamma_p \mathbf{E}.$$

Then $\Delta \rightarrow -k^2$ and $\partial_t \rightarrow i\omega$, so

$$-k^2 + \frac{\omega^2}{c^2} = \mu_0 i \omega \gamma_p.$$

Using $\gamma_p = -ine^2/(m_e \omega)$,

$$-k^2 + \frac{\omega^2}{c^2} = \mu_0 \frac{ne^2}{m_e}.$$

Since $\mu_0 = 1/(\varepsilon_0 c^2)$,

$$k^2 = \frac{\omega^2 - \omega_p^2}{c^2}, \quad \omega_p^2 = \frac{ne^2}{m_e \varepsilon_0}.$$

Final answer.

$$\omega^2 = \omega_p^2 + c^2 k^2, \quad \omega_p^2 = \frac{ne^2}{m_e \varepsilon_0}.$$

Question 16 – Plasma cutoff frequency

For propagation, the wave number k must be real. Since

$$k^2 = \frac{\omega^2 - \omega_p^2}{c^2},$$

we need

$$\omega > \omega_p.$$

If $\omega < \omega_p$, then $k^2 < 0$, k is imaginary, and the wave is evanescent: it cannot propagate through that plasma region.

The cutoff frequency is

$$f_p = \frac{\omega_p}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{ne^2}{m_e \varepsilon_0}}.$$

For $n = 1 \times 10^{11} \text{ m}^{-3}$,

$$\omega_p = \sqrt{\frac{(10^{11})(1.60 \times 10^{-19})^2}{(9.11 \times 10^{-31})(8.85 \times 10^{-12})}} \simeq 1.8 \times 10^7 \text{ rad s}^{-1}.$$

Therefore

$$f_p = \frac{1.8 \times 10^7}{2\pi} \simeq 2.8 \times 10^6 \text{ Hz} \simeq 3 \text{ MHz}.$$

Final answer.

$$f_p \simeq 3 \text{ MHz} \quad \text{for} \quad n = 1 \times 10^{11} \text{ m}^{-3}.$$

Waves below this frequency cannot propagate through such a dense plasma.

Question 17 – Interpreting the 2.2 MHz radar map

A radar echo returns when the wave is reflected. In this plasma model, reflection happens when the local plasma frequency is at least the radar frequency.

For $f = 2.2 \text{ MHz}$, the electron density needed for cutoff is

$$n_{\text{cut}} = \frac{m_e \varepsilon_0 (2\pi f)^2}{e^2}.$$

Numerically,

$$n_{\text{cut}} \simeq 6 \times 10^{10} \text{ m}^{-3}.$$

This is close to the sprite characteristic density $1 \times 10^{11} \text{ m}^{-3}$ found in Question 16.

Figure 6 shows a strong reflecting region around roughly 90 km to 105 km, which is expected for the lower ionosphere. It also shows narrow transient echo structures extending downward to around 50 km to 80 km. Those localized structures are consistent with extra ionization caused by high-altitude electrical discharges.

Final answer. The 2.2 MHz radar reflects from regions with $n_e \gtrsim 6 \times 10^{10} \text{ m}^{-3}$. The transient columns in Figure 6 are consistent with localized ionization produced by sprite-related discharges.

5 Formation avalancheuse de courant

Question 18 – Survival equation before collision

Consider a slice between x and $x + dx$. Its volume is

$$\Sigma dx.$$

If the molecule number density is n_{mol} , the number of molecules in the slice is

$$n_{\text{mol}} \Sigma dx.$$

Each molecule presents an effective collision cross-section S_{eff} . The total effective area in the slice is therefore

$$n_{\text{mol}} \Sigma dx S_{\text{eff}}.$$

Dividing by the total section Σ , the probability that an electron collides in this slice is

$$n_{\text{mol}} S_{\text{eff}} dx.$$

If $N(x)$ electrons have reached x without collision, the number lost in the next slice is

$$dN = -N(x)n_{\text{mol}}S_{\text{eff}} dx.$$

So

$$\frac{dN}{dx} = -n_{\text{mol}}S_{\text{eff}}N.$$

Solving with $N(0) = N_0$,

$$N(x) = N_0 \exp(-n_{\text{mol}}S_{\text{eff}}x).$$

Final answer.

$$\frac{dN}{dx} = -n_{\text{mol}}S_{\text{eff}}N, \quad N(x) = N_0 \exp(-n_{\text{mol}}S_{\text{eff}}x).$$

Question 19 – Mean free path

The survival law has the form

$$N(x) = N_0 e^{-x/\ell_{pm}}.$$

Comparing with the result of Question 18,

$$e^{-x/\ell_{pm}} = e^{-n_{\text{mol}}S_{\text{eff}}x},$$

so

$$\ell_{pm} = \frac{1}{n_{\text{mol}}S_{\text{eff}}}.$$

This is also the average distance before a collision for an exponential free-path distribution.

Final answer.

$$\ell_{pm} = \frac{1}{n_{\text{mol}}S_{\text{eff}}}.$$

Question 20 – Mean free path at 60 km

The value $S_{\text{eff}} \sim 1 \times 10^{-20} \text{ m}^2$ is reasonable because a molecule has a size of order $1 \times 10^{-10} \text{ m}$; an area is therefore of order

$$(10^{-10})^2 = 10^{-20} \text{ m}^2.$$

Use the ideal gas law in number-density form:

$$n_{\text{mol}} = \frac{P}{k_B T}.$$

At 60 km, the statement gives $P_3 = 20 \text{ Pa}$ and $T_3 = 260 \text{ K}$, so

$$n_{\text{mol}} = \frac{20}{(1.38 \times 10^{-23}) \times 260} \simeq 5.6 \times 10^{21} \text{ m}^{-3}.$$

Then

$$\ell_{pm} = \frac{1}{(5.6 \times 10^{21})(10^{-20})} \simeq 1.8 \times 10^{-2} \text{ m}.$$

Final answer.

$$\ell_{pm} \simeq 2 \times 10^{-2} \text{ m} \simeq 2 \text{ cm.}$$

Question 21 – Can a resting electron ionize nitrogen?

The electric field is $E \simeq 100 \text{ V m}^{-1}$. Over one mean free path, the potential difference is

$$\Delta V = E\ell_{pm} \simeq 100 \times 0.018 \simeq 1.8 \text{ V.}$$

An electron crossing a potential difference of 1.8 V gains 1.8 eV. The ionization energy of nitrogen is

$$\mathcal{E}_i \simeq 16 \text{ eV.}$$

So the typical energy gained before a collision is far too small.

The distance needed to gain 16 eV is

$$x_i = \frac{16 \text{ V}}{100 \text{ V m}^{-1}} = 0.16 \text{ m.}$$

In units of the mean free path,

$$\frac{x_i}{\ell_{pm}} \simeq \frac{0.16}{0.018} \simeq 9.$$

The probability of travelling that far without collision is roughly $e^{-9} \sim 10^{-4}$. Such a long free path is possible, but rare.

Final answer. A resting electron typically gains only about 2 eV before collision, much less than 16 eV. It will not normally ionize N_2 ; only rare paths of about nine mean free paths, with probability $e^{-9} \sim 10^{-4}$, could do it.

Question 22 – Why 1 MeV cosmic electrons are relativistic

The rest energy of an electron is

$$m_e c^2 \simeq 0.511 \text{ MeV.}$$

A kinetic energy of 1 MeV is about twice the rest energy. Therefore the non-relativistic approximation cannot be valid.

The Lorentz factor is

$$\Gamma = 1 + \frac{K}{m_e c^2} \simeq 1 + \frac{1}{0.511} \simeq 2.96 \simeq 3.$$

Then

$$\frac{v}{c} = \sqrt{1 - \frac{1}{\Gamma^2}} \simeq 0.94.$$

Final answer. A 1 MeV electron has $\Gamma \simeq 3$ and $v \simeq 0.94c$, so it is necessarily relativistic.

Question 23 – Cascade avalanche explanation

A high-energy primary electron can ionize a nitrogen molecule during a collision. That collision releases a secondary electron. After the first ionization, there are more free electrons than before. These electrons can themselves be accelerated and produce more ionizations.

In the simplest model, each stage roughly doubles the number of electrons:

$$1 \rightarrow 2 \rightarrow 4 \rightarrow 8 \rightarrow \dots$$

This is exponential growth. The process is called an avalanche because each generation creates more particles able to create the next generation.

The avalanche cannot grow forever. It can be limited by energy losses, recombination, attachment to neutral molecules, or screening of the electric field by the newly produced charges. During the process, excited molecules can emit visible light, which contributes to the sprite glow.

Final answer. A cascade avalanche is a chain reaction of impact ionizations. One energetic electron creates secondary electrons, which create more ionizations, leading to approximately exponential growth of electron number and current.

Question 24 – Avalanche stage number

We want the average electron density to reach

$$n = 1 \times 10^{11} \text{ m}^{-3}.$$

Let $\mathcal{N}$ be the number of avalanche stages. In a doubling model, the number of electrons after $\mathcal{N}$ stages is approximately

$$2^{\mathcal{N}}.$$

The problem states that the electrons are contained in a characteristic volume

$$(\mathcal{N}\ell_{pm})^3.$$

Using $\ell_{pm} \simeq 2 \text{ cm} = 2 \times 10^{-2} \text{ m}$ from Question 20,

$$n \simeq \frac{2^{\mathcal{N}}}{(\mathcal{N}\ell_{pm})^3}.$$

Set this equal to $1 \times 10^{11} \text{ m}^{-3}$:

$$10^{11} = \frac{2^{\mathcal{N}}}{\mathcal{N}^3(2 \times 10^{-2})^3}.$$

So

$$2^{\mathcal{N}} = 10^{11}\mathcal{N}^3 \times 8 \times 10^{-6} = 8 \times 10^5 \mathcal{N}^3.$$

Equivalently,

$$2^{\mathcal{N}-3} = 10^5 \mathcal{N}^3.$$

Multiplying by 10^{-9} gives

$$10^{-9} \times 2^{\mathcal{N}-3} = 10^{-4} \mathcal{N}^3.$$

This is exactly the equality represented by the two curves in Figure 7. This check also explains why the graph is calibrated for $\ell_{pm} \simeq 2 \text{ cm}$: it is what converts the density equation into the two plotted functions.

At $\mathcal{N} = 35$,

$$10^{-9}2^{35-3} \simeq 4.29, \quad 10^{-4}35^3 \simeq 4.29,$$

so the intersection is genuinely near

$$\mathcal{N} \simeq 35.$$

The corresponding length scale is

$$\mathcal{N}\ell_{pm} \simeq 35 \times 2 \text{ cm} \simeq 0.7 \text{ m}.$$

Final answer.

$$\mathcal{N} \simeq 35,$$

so to one significant digit the avalanche requires about 4×10^1 stages.

Final checked results

Quantity	Checked value
Pic du Midi–Alps distance order	$d \sim 500 \text{ km}$
Curvature height over that distance	$h \sim 20 \text{ km}$
Gravity at ISS altitude	$G_{\text{ISS}} \sim 9 \text{ m s}^{-2}$
ISS speed	$v_{\text{ISS}} \sim 8 \text{ km s}^{-1}$
ISS period	$T_{\text{ISS}} \sim 90 \text{ min}$
Atmospheric scale height	$H \sim 9 \text{ km}$
Plasma cutoff for $n = 10^{11} \text{ m}^{-3}$	$f_p \sim 3 \text{ MHz}$
Radar cutoff density at 2.2 MHz	$n_{\text{cut}} \sim 6 \times 10^{10} \text{ m}^{-3}$
Mean free path at 60 km	$\ell_{pm} \sim 2 \text{ cm}$
Energy gained over one mean free path	$eE\ell_{pm} \sim 2 \text{ eV}$
Avalanche stage number	$\mathcal{N} \sim 35 \sim 4 \times 10^1$